

Financing investment: Evidence from Russian public companies

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JEL codes: E22, G32, L32

Keywords: investment, financial constraints, soft budget constraints, debt overhang, state-owned enterprises

The paper considers the factors of investment activity of Russian companies before and after the global financial crisis. Given the tightened access to financing and at the same time the observed continued growth of the corporate debt we investigate the role of financial constraints and the role of debt overhang, which is identified using a new indicator proposed in the paper. Empirical evidence demonstrates substantial differences in the investment behavior of private companies and companies with government participation in the capital. State-owned companies have soft budget constraints and are not subject to debt overhang. This may contribute to state companies crowding private companies out of the loanable funds market.

1. Introduction

After the global financial crisis Russian macroeconomic dynamics changed dramatically. The period of recovery growth which followed the transformational recession ended, and a change in external financial and economic conditions is to provide additional dampening effect on economic growth in Russia.

Bank credit in Russia before the years of 1998-2000 was very small. Banks worked primarily as depositors and speculative investors. As a result, firms have been forced to finance many of investment projects themselves. The subsequent development of the Russian banking sector in the 2000s and improved access to credit is reflected in the fact that less than 50% of the funds that Russian enterprises use as sources of investment financing are their own and more than 30% are borrowed (20% are budgetary funds).

Before the last global crisis (until 2008), financial conditions were soft and Russian companies actively used debt financing – the debt of non-financial corporations to GDP increased by almost 1.5 times from 2005 to 2008 (Fig. 1) – and investments grew quite rapidly (Fig. 2).

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Monetary tightening and reduced access to external financing, together with the expected worsening economic outlook led to a reduction in the debt burden during the crisis and post-crisis period, but starting from 2012 corporate debt was growing again, and now has exceeded the pre-crisis peak. The dynamics of investment was quite different: post-crisis recovery gave way to a new period of falling, the ratio of investment to GDP decreased² (Fig. 2).

Thus, in the post-crisis period in Russia we observed both the tightening of macroeconomic financial conditions and the continued growth of corporate debt. On the one hand, it gives reason to expect that companies became much more financially constrained, and on the other – that the impact of financial constraints were partly offset by other effects, which resulted in the growth of the total debt in spite of the tightening and worse economic prospects growth.

In our study we test the hypothesis that one of the reasons for the investment stagnation is high debt burden of Russian companies – *debt overhang* - however, not for all of them, but for private ones, while companies with state participation are practically not subject to the influence of this phenomenon and keep on increasing the debt.

The concept was introduced in the classic work by Myers (1977), who analyzed the impact of the company's debt on the investment strategy as a result of the conflict of interests between company owners and its creditors. The owners (and managers representing them) of the company with a high debt burden may have little incentive for the implementation of investment projects, since the large share of profits from the project will depart to the creditors (perhaps even completely – in case of bankruptcy of the company). Thus, other things being equal, a company with higher debt will invest less than a firm with low debt. Not only shareholders do not want to implement projects, since a significant share of the benefits would go to existing lenders, but also attracting new lenders may then be difficult for the same reason. In addition one of the channels through which debt overhang has an impact on the investment is the reduction of company's free cash flow resulting from the need to service the debt (Jensen, 1986).

Debt overhang may impose a limit on the threshold yield of the investment project. If the profits of the project are less than the amount necessary for the implementation of all payments on existing debt and payments to new investors and lenders, the project will not be implemented, even if it is profitable. Demonstration of this idea in the form of a simple model is given by Lamont (1995).

The existence of such thresholds do not play an important role in a fast-growing economy (when the return on investment is high), and can be crucial during stagnation. For this reason the same value of accumulated debt can generate or not debt overhang problem in different economic

² Investment in non-financial assets.

conditions. That is why, as noted in (Lamont, 1995), exacerbation of the problem and its impact on investment and macroeconomic dynamics should be observed during crisis and post-crisis periods. It was the case in Russia after the global financial crisis: the transition from a rather rapid growth to stagnation.

Given the fact that by the time of the crisis the Russian corporate sector's both domestic and foreign debt had been high enough, we expect that the debt overhang began to play a more prominent role in the dynamics of corporate investment.

However it is well-known that in Russia the government plays a substantial role in the economy and is a shareholder in many large companies. Companies associated with the government have both formal and informal preferences, guarantees and may be eligible for state support in financing and refinancing, so they are expected to have a soft budget constraints. Analyzing the differences in the behavior of private companies and companies with state participation in the capital (hereinafter for simplicity we call them state-owned companies, although we do not necessarily mean a controlling stake but any share held by the government), we test this hypothesis. In addition, there're reasons to expect that access to debt financing is also easier for state-owned companies.

In Russia the largest banks are state-owned ones and this is one of the most important channels through which the state can influence corporations. The role of state-owned banks may be significant regarding their sometimes close relationships with major borrowers, many of which are companies with state participation. However, in Russia, the government traditionally supports mining industries. For other companies-potential borrowers such mechanisms of capital allocation can have negative consequences, since it embarrasses their access to financing. The latter may lead to a slowdown in investment in other sectors which provide innovative development and stability of Russian economic growth.

An important aspect in this context is the degree of development of national financial markets. The researchers empirically confirmed the hypothesis that financial development is significant for economic growth ((Levine, Zervos, 1998), (Beck, et al., 2000), (Beck, Levine, 2004)). In countries with more developed financial markets sectors with a high demand for debt financing, which are mostly high-tech industries, are growing relatively faster (Rajan, Zingales, 1998). In Russia, national financial markets are still rather far from to be called developed, and active participation of the state could aggravate the access to finance for those industries and firms which it does not directly support.

Our hypothesis is as follows. Under limited financial resources, companies related to the government, may "crowd out" other companies out of the capital market. The negative consequences for the economy stem from the reduction in investment in industries that are not

supported by the state, which in the current environment is underinvestment in non-primary sectors of the economy. This in turn leads to the preservation of traditional economic structure and prevent its diversification. Weak diversification of the economy, first, strengthens cyclical fluctuations, and secondly, lowers growth rates in the long run (see, eg., (Lederman, Maloney, 2007)).

One evidence of this is given in Fig. 3, which shows the share of non-financial corporations, related to the so-called *public sector in the extended definition* in the total foreign corporate debt. According to the classification of the Bank of Russia, "*the external debt of the public sector in the broader definition encompasses the external debt of the non-bank corporations ... in which government and the central bank directly or indirectly own 50 percent or more equity or control them in any other way*". The share of these companies is quite high – the average for the last ten years is more than 30%. Remarkably, this ratio is countercyclical: reaching the "bottom" in the soft pre-crisis conditions, after the crisis it has been growing steadily. It's noteworthy that in this case we are talking only about companies in which the government has a controlling stake, ie excluding those in which its participation is less than 50%. Including the latter would obviously make the ratio even higher.

Countercyclical state-owned enterprises' share in the total external debt favors our hypothesis of relatively soft budget constraints, and in addition, gives rise to testing the hypothesis that for such firms the problem of debt overhang is not so acute. If so, under the conditions of limited financial resources state-owned companies may "crowd out" private companies from the capital market.

The issue of financing investment under limited financial resources became particular relevant for Russian enterprises in 2014: starting this year, due to political reasons, the international capital markets are closed for most Russian companies. The closure of foreign markets and other economic sanctions have led to a sharp weakening of the ruble and the tightening of domestic monetary conditions. Fig. 4 shows the index of bank lending tightness (BLT), which is calculated by the Bank of Russia, and its components: in 2014 there has been a consistent and strong tightening of credit conditions, which continued into early 2015. The closure of global markets led to the fact that 2014 was the first year when non-financial corporates reduced their external debt (even taking into account the effect of national currency devaluation), replacing it by internal debt. However, this has not led to a reduction in the debt burden as it is: the debt service ratio for 2014 rose sharply (Bank of Russia, 2015), surpassing the previous peak of the crisis 2009. All this is evidence for the debt overhang to worsen.

In the new financial environment for Russian economy starting from 2014 (due to political and economic sanctions), an indirect confirmation of the "crowding out" effect is the increase in lending to large companies together with the rapid decline in lending to small and medium-sized

businesses (Bank of Russia, 2015). While here we are talking about changes in the structure of lending depending on the size of firms, this may be due to the participation of state, because government is a shareholder in many of the biggest Russian corporations.

In this paper we analyze the factors of investment of private companies and companies with government participation before and after the global financial crisis. First, we test the hypothesis of the existence of debt overhang for Russian companies using a proposed new indicator to identify the presence of this phenomenon. Second, we examine the differences in the behavior of private and state-owned companies to test the hypothesis of the existence of soft budget constraints for the latter.

Our results suggest that the problem of debt overhang took place before the crisis, but afterwards increased significantly: before 2009 the debt burden has almost no disincentive effect on investment, but after – we find quite significant negative impact. In addition it appears that state-owned companies are subject to much less debt overhang. In general, financial constraints for companies with government participation are much weaker than for private ones.

Our paper proceeds as follows. In Section 2 we discuss related literature as well as offer an empirically testable hypothesis to identify the debt overhang. Section 3 describes the data and estimation method. The results of econometric estimation are presented in Section 4. Section 5 concludes.

2. Related literature. Empirically testable hypothesis

Modigliani-Miller theorem (Modigliani, Miller, 1958) and based on it neoclassical theory of investment (Hall, Jorgenson, 1967) assumes perfect capital markets, universe interest rate at the macroeconomic level and independence of decisions about real variables (investments, etc.) from financial ones (capital structure, etc.), which implies that investment depends solely on investment opportunities expressed in the value of Tobin's Q (Tobin, 1969). But empirical evidence suggests that capital markets are imperfect, internal and external financial resources are not perfect substitutes. Many studies consider determinants of corporate investment behavior and focus on finding out the role of financial constraints (eg., Fazzari et. al, 1988; Schiantarelli, 1996; Bond et. al, 2003). It was shown that free cash flow of the company is important. The subject of interest was investment's dependence on financial variables, among others the impact of leverage. As a result common place in the literature on corporate finance is that *ceteris paribus* those firms with more cash and less debt invest more (Handbook of the Economics of Finance, 2003). The negative relationship between debt and investment became a confirmation of the hypothesis of debt overhang, first described by Myers (1977).

While some studies suggested the integration of debt overhang in macroeconomic models (Lamont, 1995; Gertler, 1992; Occhino, Pescatori, 2014; Chen, Manso, 2014), an important issue is empirical testing of this phenomenon presence. A number of authors use regression relationship between investment (normalized to assets) and leverage as a measure of the debt overhang (eg., Gallego, Loayza, 2004; Aivazian, 2005; Firth et al., 2008). Another method proposed in (Hennessy, 2004) is using recovery ratios, the calculation of which for Russia is difficult because of the lack of necessary data. Hennessy also notes that leverage is not a good proxy for debt overhang.

We offer another form of empirically testable hypothesis. When constructing a hypothesis, we rely on the theory of financially constrained investment. This theory is illustrated in Fig. 5a (Hubbard, 1998), which shows the curves of supply and demand for investment. The position of the demand curve is determined by investment opportunities. The supply curve in perfect conditions must be horizontal, defined by a single interest rate in the market (r), however due to the asymmetry of information it has a positive slope and a break, both because the lenders require a premium for investing their funds. The position of the break is determined by the firm's own financial resources (W – worth). The slope depends on the degree of asymmetry of information – information costs³. As a result, investment I_0 and I_1 are less than optimal value K^* .

We assume that the slope of the supply curve, that is the size of the risk premium, is positively determined by the company's debt burden: greater present debt makes the attraction of new debt more risky and costly. Greater leverage makes the curve steeper. Now integrating this idea into the described scheme, one can trace what is happening to the investment along the growth of debt (leverage), other things being equal. This is depicted in Fig. 5b. The growth of leverage from lev_0 to lev_1 rotates the supply curve to the left (holding constant own funds W) and reduces investment from I_0 to I_1 , while the value of new debt raised (to finance these capital expenditures) d decreases faster (from d_0 to d_1), and the ratio I/d increases (I_1/d_1 is greater than I_0/d_0), which means that companies have to finance most of the investments themselves. Therefore, we can write:

$$\frac{I_t}{d_t} = \alpha \frac{D_{t-1}}{A_{t-1}}, \quad (2)$$

where D_{t-1}/A_{t-1} – leverage, ie the ratio of debt to assets (end of period $t-1$, ie, beginning of period t), and the coefficient α is positive, and the greater α the faster the supply curve turns to the left, becoming steeper. Thus the greater α corresponds to “more” debt overhang.

Then, in order to integrate the ratio into the estimation equation, which we will assess using traditional dependence of investment normalized by assets, rewrite (2) in the following way:

³ Более подробное описание графика и теории, которую он иллюстрирует, см. в работе (Hubbard, 1998).

$$\frac{I_t}{A_{t-1}} = \alpha \left[\frac{d_t}{A_{t-1}} \times \frac{D_{t-1}}{A_{t-1}} \right]. \quad (3)$$

We built empirically testable hypothesis which we shall use in this paper. Coefficient α will test the hypothesis on the extent of debt overhang.

Results obtained in the empirical studies of debt overhang indicate its importance. In (Kalemli-Ozcan et. al, 2015) the authors study the latest debt crisis in the Eurozone and conclude that the high indebtedness of European firms become one of the major reasons for the decline of investment. Another interesting result is that firms which have a high proportion of short-term debt are particularly affected. The same conclusion is made in (Duchin et. al, 2010), who studied the impact of the global financial crisis of 2008 on the investment determinants.

The existence of debt overhang is confirmed for US firms' data (Hennessy, 2004). This study attempted to find the differences in effect size for different groups of companies – in this case, depending on the investment rating – but they didn't. Our work in this regard is close because we also want to identify the companies for which debt overhang is less of an issue than for others. In our study these differences were found – between private companies and companies with government participation.

These differences indicate soft budget constraints for state-owned companies. Soft budget constraints are generated as a result of a wide variety of ways to support companies by the state, ranging from direct subsidies and government loan guarantees and ending with the informal support: negotiations with creditors, confidence in the bailout in the case of serious financial difficulties (Kornai et. al, 2003). The presence of this phenomenon in the investment function was also obtained by the authors who have studied the corporate sector in Bulgaria (Budina et. al, 2000), although they divide companies not by ownership structure, but according to their size – soft budget constraints appeared to be typical for large companies. This result echoes the substitution of loans for small and medium-sized enterprises by loans to large ones, which we observed in Fig. 6. In our study testing this hypothesis is difficult because we are dealing with a sample where all companies are large.

3. Data and method of estimation

We have formed a database of more than 100 Russian public companies, for the period from 2000 to 2014. Source of data is Thompson Reuters. The database includes the financial indicators of corporations (from balance sheet, income statement, cash flow statement), a number of market indicators (including market capitalization of the company), as well as information about government participation in capital. Thus, estimation is performed on annual panel data.

The first specification which we want to assess is the equation without heterogeneity in the ownership structure of companies:

$$inv_{i,t} = \beta_0 + \beta_1 \cdot inv_{i,t-1} + \beta_2 \cdot qtobin_{i,t-1} + \beta_3 \cdot cf_{i,t-1} + \beta_4 \cdot debto_{i,t} + \mu_t + \varepsilon_{i,t}, \quad (4)$$

where $inv_{i,t}$ – investment in non-financial assets for company i during the period t , normalized by the assets, beginning of period t (ie, end of period $t-1$), I_t/A_{t-1} ;

$qtobin$ – Tobin's Q, calculated as the market value of the company to the book value of its assets, where the market value is obtained as the sum of market capitalization and debt;

cf – free cash flow (operating income);

$debto$ – variable for the debt overhang, introduced in Section 2: the product of new attracted debt to assets (dch_t) and leverage (lev_{t-1}):

$$\frac{d_t}{A_{t-1}} \times \frac{D_{t-1}}{A_{t-1}};$$

μ – period fixed effects;

ε – error.

The model is estimated separately for two periods: 2000-2008 and 2009-2014. The first period is regard as the pre-crisis, since the Russian market crisis began in the autumn of 2008 (after Lehman Brothers collapse), but we assume that the major investment decisions of corporations for the year of 2008 by this point of time (autumn of 2008) had been already taken. The second time period includes the crisis in Russian market (2009) and post-crisis period (including 2014 – the year of sharply tightened financial conditions).

The second specification we estimate is the equation which takes into account the effect of the government share in the company's capital:

$$inv_{i,t} = \beta_0 + \beta_1 \cdot inv_{i,t-1} + \beta_2 \cdot qtobin_{i,t-1} + \gamma_2 \cdot qtobin_{i,t-1} \times state_{i,t} + \beta_3 \cdot cf_{i,t-1} + \gamma_3 \cdot cf_{i,t-1} \times state_{i,t} + \beta_4 \cdot debto_{i,t} + \gamma_4 \cdot debto_{i,t} \times state_{i,t} + \mu_t + \varepsilon_{i,t}, \quad (5)$$

where $state$ is multiplicative dummy variable: 1, if the state has a stake in the company, and 0, otherwise.

So we are assessing all companies in one sample but take into account the differences in the coefficients for each of the groups of companies.

We use the generalized method of moments (GMM) for dynamic panel data models developed by Arellano and Bond (1991). The dynamic panel data is characterized by the presence of the lagged component. One feature of such models is the presence of individual effects that are correlated with an error. This correlation leads to the fact that estimating this kind of models by least squares provides inconsistent estimates. Removing individual effects by building “within” estimates does not solve the problem because it does not eliminates the correlation with the error (estimates are also inconsistent). Least squares estimates are biased upwards, and “within” –

downwards (Bond, 2002). The generalized method of moments for dynamic panel data models solves this problem by taking first differences of dependent variables and assessment models in this form.

In addition it solves the problem of the lack of strict exogeneity of explanatory variables by using instrumental variables. In our case, we can talk about weak exogeneity. Weak exogeneity means that future innovations of dependent variable does not affect current values of the explanatory variables. The method is based on the use of internal tools – several consecutive lags of explanatory variables in levels (ie levels are tools for differences). We use two lags for each variable. The consistency of the obtained estimates is ensured by the condition of validity of instruments, which can be verified by Sargan test for over-identifying restrictions (J-statistics) (the null hypothesis is that the instruments are valid).

4. Results of estimation

Before we go directly to the results of the econometric estimation, let's look at descriptive statistics. Tab. 1 shows statistics for a sample divided into two periods of interest, but without separation by ownership. First of all, investment decreased dramatically: from 15% to 9.5% with respect to the assets (the median indicator declined from 12% to 8%). Indicators of Tobin's Q and free cash flow also decreased. At the same time the average leverage increased by almost half. But in spite of this growth, our indicator of debt overhang (*debto*) declined due to more powerful reduction of *dch*, that is, companies have attracted fewer new loans on average. We remind that the decline in the indicator *debto* corresponds to exacerbating the debt overhang.

The estimation results of specification (4) for each of the periods are presented in Tab. 2⁴. The situations before and after the crisis differ considerably in terms of the factors of investment. Before the crisis, companies did not experience severe limitations in terms of financing, which made investments depending on Tobin's Q without significant dependence on free cash flow. After the crisis emerged the consequences of the tightening of financial conditions in the economy. First, the dependence on Tobin's Q fell sharply, and secondly, free cash flow became a significant factor. The phenomenon of debt overhang was present in both the periods under review, but in the post-crisis one its influence grew.

Now consider what would change if we take into account the heterogeneity of the companies in terms of government participation in their capital. First, we analyze the descriptive statistics presented in Tab. 3. It can be seen that the behavior of the companies turned out

⁴ Unfortunately, despite the large number of companies and a long time period, the data contain many gaps, and the lag structure of our variables imposes additional restrictions, that is why we have a relatively small number of observations in the period from 2000 to 2008.

reasonably different (Fig. 6 also). While investments of state-owned companies did not decrease much (from 12.7% to 10.8% to assets), investments of private companies suffered significantly – almost two-fold reduction, from 16.5% to 8.9%. Tobin’s Q for private companies both before and after the crisis was higher than for the state, but this difference is a subject for special study, as it can possibly be explained by sectoral differences. In general, this study does not take into account sectoral differences, though, as we expect, they can play a prominent role, and it is a promising line of research.

Remarkably, the profit of the companies does not differ depending on the type of ownership, which means that all other differences between firms that we are observing, are not determined by possible differences in their profitability (which actually is not present).

The most interesting analysis of the differences is considering debt variables. For private companies measure the average change in debt (*dch*) fell sharply after the crisis (from 12.7% to 5.4% of the assets), but for the state, on the contrary, increased (from 5.7% to 6.7%). This situation is quite abnormal, given that tightening monetary conditions should have led to a reduction in new borrowings everywhere. We consider this fact as an additional signal in favor of the existence of soft budget constraints: state-owned companies are able to borrow in shrinking funds markets.

Due to the growth of new attracted debt of state-owned companies the indicator *debt* also increased, what would mean an average reduction of debt overhang for state-owned companies after the crisis. We test this hypothesis and other hypotheses about the determinants of investment for companies with different ownership structure, considering the specification (5). The estimation results are shown in Tab. 4. The first column for each period are the regression coefficients, however, since we are dealing with deviations, which are the result of the division of the state-owned companies group, in the second column we put the resulting coefficients for state-owned companies ($\beta_j + \gamma_j, j=2, 3, 4$) and then all the outcomes we are interested in are in bold.

We found that before the crisis there were almost no differences in the behavior of private companies and state-owned companies, as there were no significant financial constraints: capital expenditures were determined mainly by investment opportunities (Tobin’s Q). Changes have occurred since 2009. Firstly, the dependence of investment on Tobin’s Q has fallen. Second, the free cash flow has begun to play a significant role for private companies, but not for the state. And third, the coefficient of the indicator of debt overhang for private companies has tripled, while for the state ones remained almost unchanged. The last two facts clearly indicate that companies with government participation are in a more favorable financial conditions and have considerably easier access to funding, that is, soft budget constraints.

In addition it is interesting to evaluate the role played by debt overhang in reducing private investment. By the end of 2008, private companies have come to the ratio of investment assets at

the level of 0.154 (15.4%). In 2009-2014, their level was 0.089 (8.9%), that is, there has been a decrease of 0,065 (6.5 percentage points). Indicator *debto* in turn during the same period fell from 0.050 to 0.016, ie 0.034. According to our regression coefficient equal to 0.339, the growth of debt overhang explains 18% of the actual reduction of the investment. The magnitude of this effect is not small to be ignored, so it can be stated that the debt overhang has significant influence on investment.

5. Conclusion

Assessing the investment functions for private and state-owned companies in the periods before and after the global financial crisis we have found that factors of investments are different either among periods and among types of firms considered. If for the pre-crisis period we have received no or little influence of financial constraints on the firms' investment decisions, in the post-crisis period these constraints began to play a leading role for private companies and still insignificant – for the state ones. A special attention is paid to the question of the impact of the accumulated debt burden. It turned out that in the tightened monetary conditions debt overhang takes place for private investment, and does not limit the company with government participation. The latter can explain the growth of corporate debt though financial conditions have tightened dramatically.

Thus we see that companies with government participation have signs of soft budget constraints and significantly easier access to finance. This result gives reason to expect that in the post-crisis environment of greatly limited financial resources, state-owned companies can crowd out private enterprises from the credit market, and thus crowd out private investment. So this effect of "quasi-public" investment can have a negative effect on the investment process as a whole.

For Russia as a country with potential to implement a catching-up economic growth strategy through the process of convergence with the global economy, as well as in the new conditions of closed global capital markets and sustainable growth slowdown (even recession), the question of funding and the factors limiting investment process is particularly relevant.

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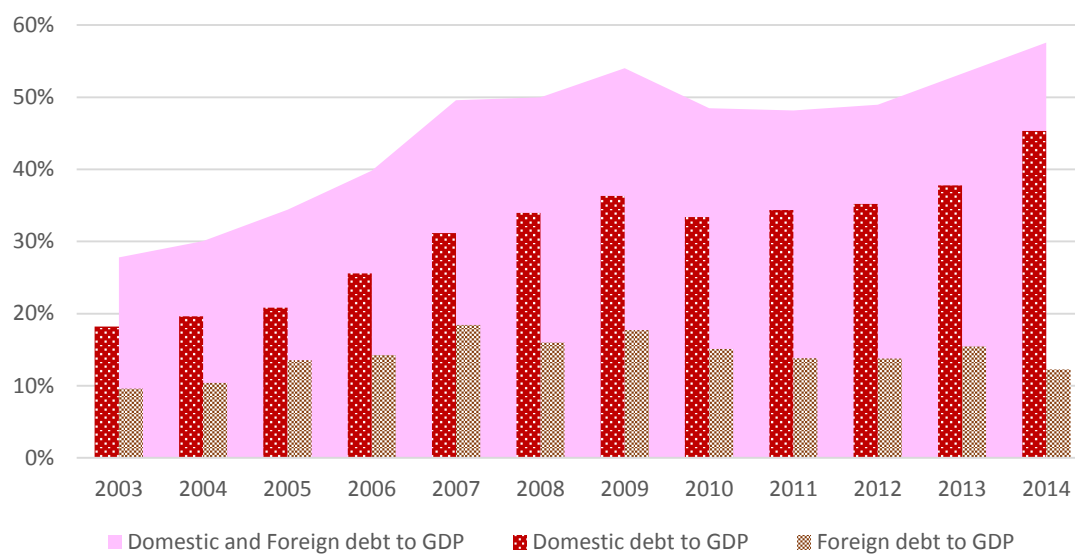
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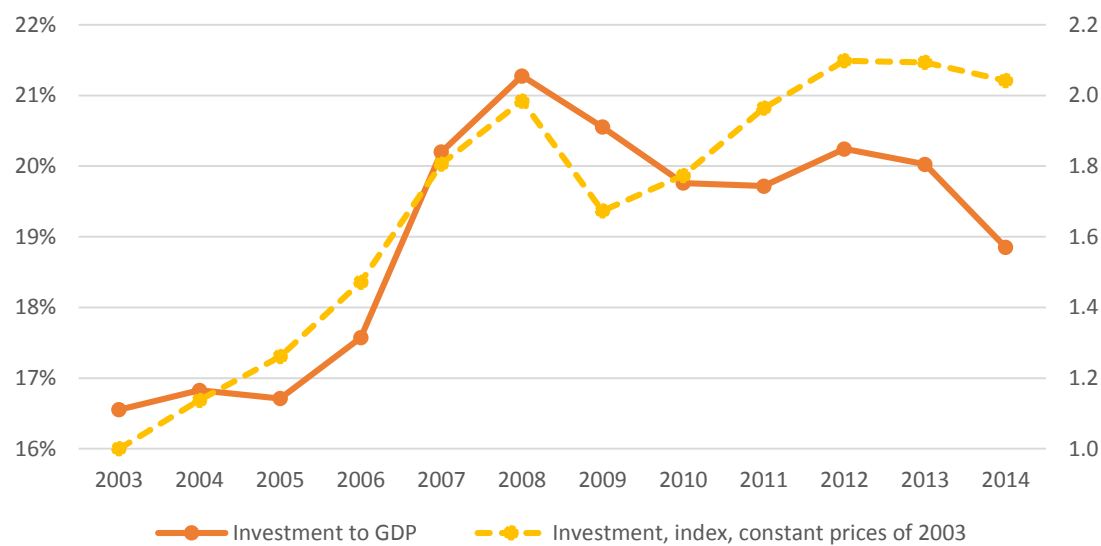
Appendix 1. Figures

Fig. 1. Non-financial Corporations' Debt



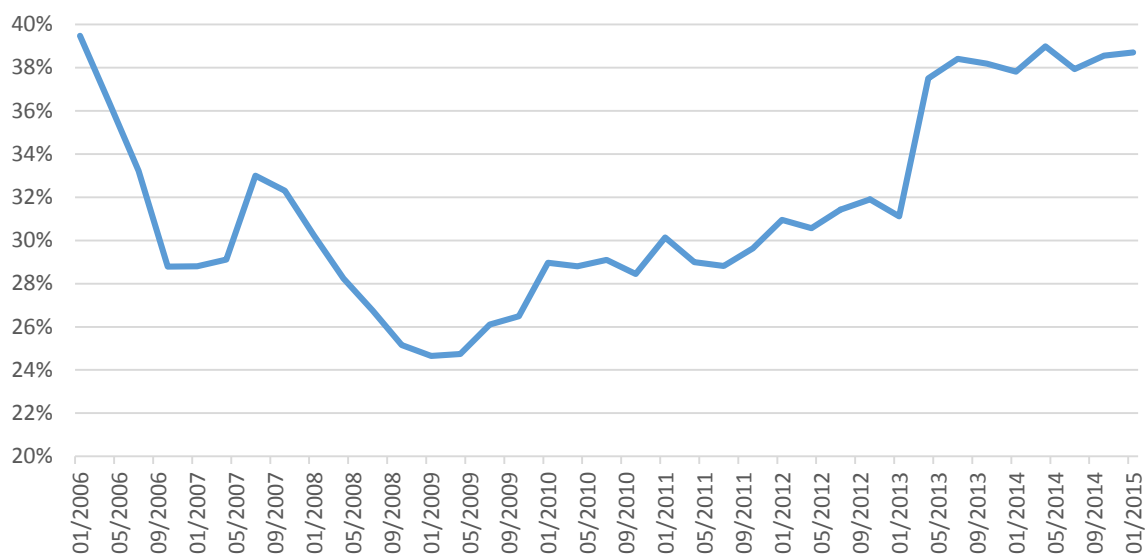
Source: Bank of Russia, Rosstat.

Fig. 2. Investment



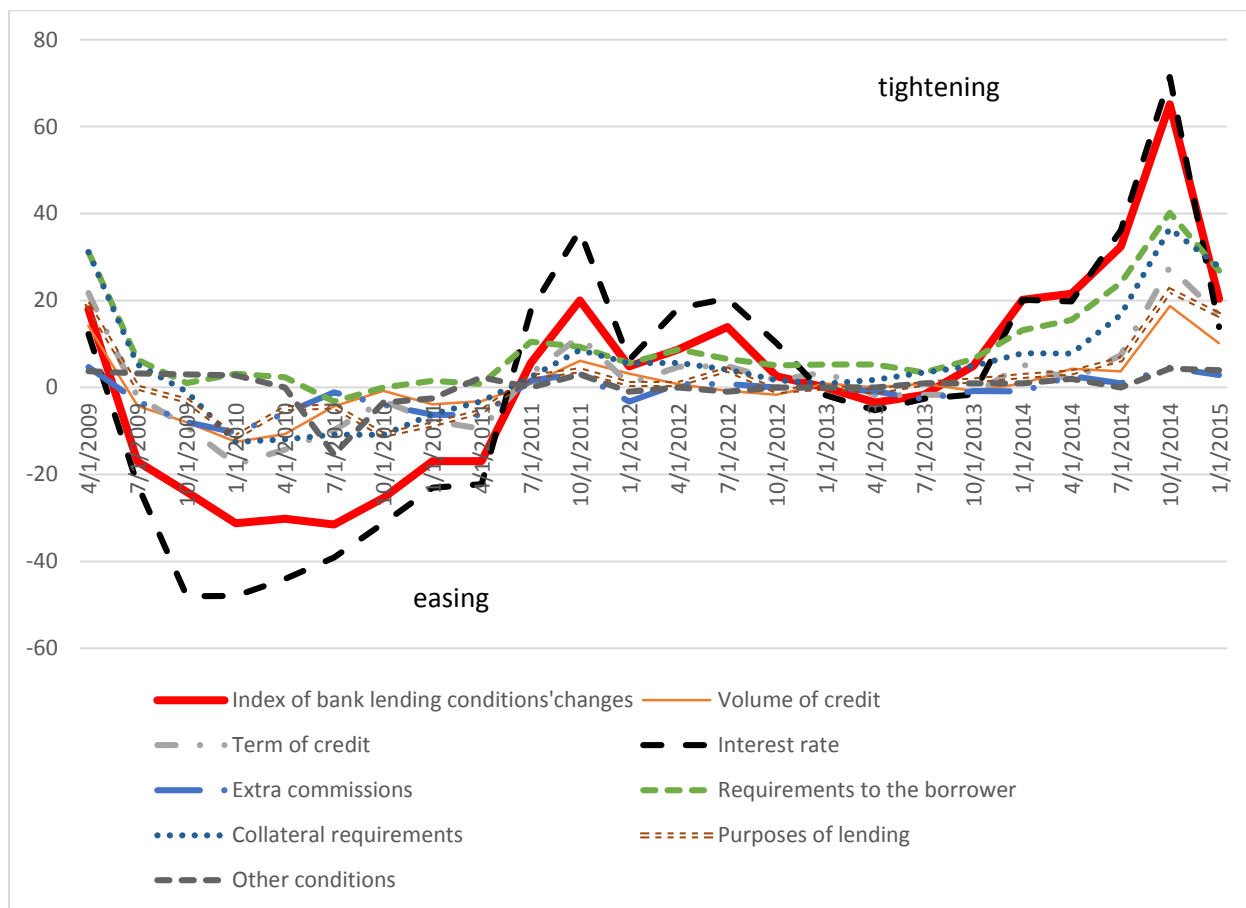
Source: Rosstat.

Fig. 3 The share of state non-financial corporations' foreign debt in the total non-financial corporations' foreign debt



Source: Bank of Russia.

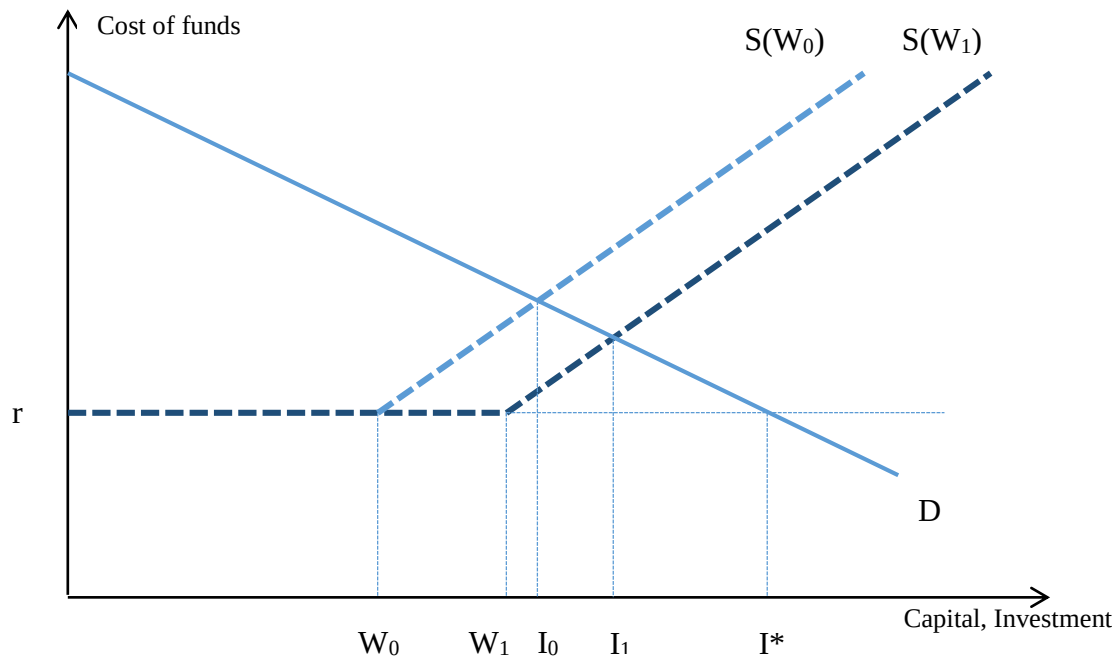
Fig. 4. Index of bank lending tightness changes and its components *



Source: Bank of Russia.

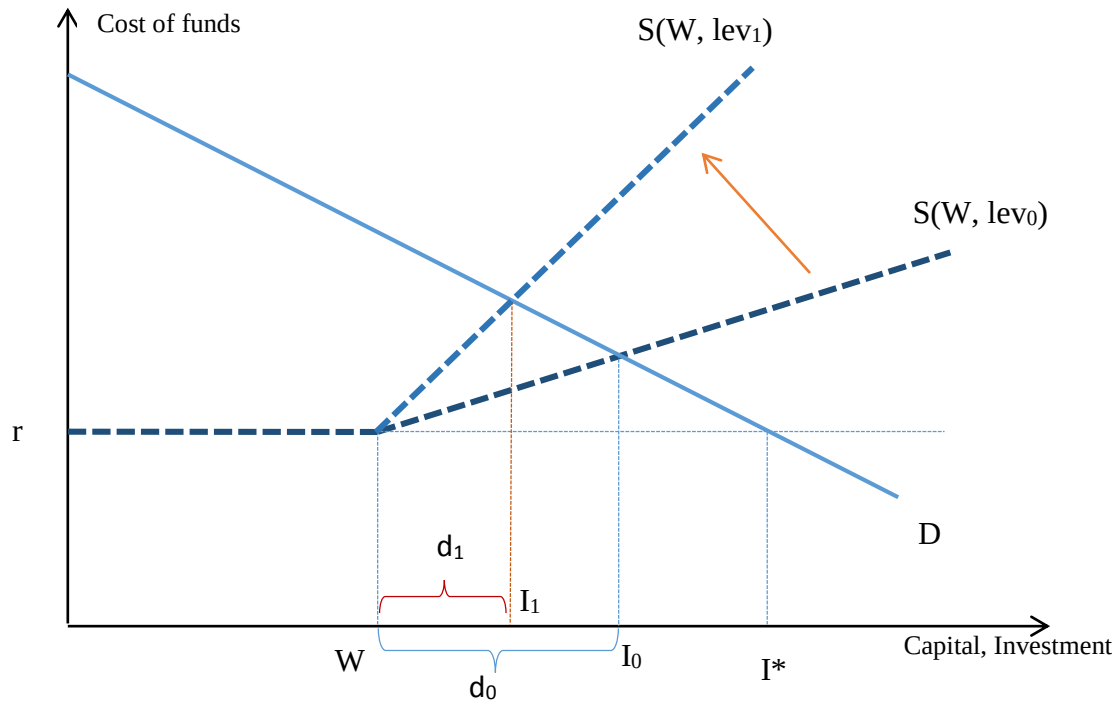
* For large enterprises. The dynamics for medium-sized and small-sized is the same

Fig. 5a. Informational Asymmetry and Underinvestment



Source: Hubbard, 1998.

Fig. 5b. Leverage and Investment



Source: (Hubbard, 1998), author.

Fig. 6. Investments and debt of private companies and companies with government participation
– before and after the crisis



Source: author's calculations.

Appendix 2. Descriptive statistics, Estimation results

Tab. 1. Descriptive statistics: before and after the crisis

	<i>inv</i>		<i>qtobin</i>		<i>cf</i>		<i>debto</i>		<i>dch</i>		<i>lev</i>	
	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14
mean	0.150	0.095	1.235	1.027	0.143	0.117	0.026	0.017	0.103	0.058	0.230	0.306
median	0.119	0.080	0.974	0.801	0.127	0.119	0.006	0.003	0.042	0.018	0.199	0.277
maximum	0.908	0.676	7.646	7.133	0.894	0.566	0.660	0.636	1.565	1.748	0.905	2.226
minimum	0.000	0.000	0.001	0.039	-0.991	-0.824	-0.213	-0.270	-0.434	-0.488	0.000	0.000
standard deviation	0.133	0.083	1.019	0.813	0.162	0.111	0.071	0.067	0.200	0.181	0.176	0.256
number of observations	464	678	415	645	542	685	480	680	483	682	602	692

Tab. 2. Estimation results, equation (4)

	2000-2008	2009-2014
<i>inv_{i,t-1}</i>	0.500*	0.217*
<i>qtobin_{i,t-1}</i>	0.097*	0.015*
<i>cf_{i,t-1}</i>	0.022	0.061*
<i>debto_{i,t}</i>	0.113*	0.267*
<i>Number of observations</i>	159	548
<i>Probability(J-statistic)</i>	0.267	0.220

* significant at 1% (5%, 10%).

Tab. 3. Descriptive statistics: private companies and companies with government participation – before and after the crisis

_s for state companies, *_ns* – for private (non-state)

	<i>inv_s</i>		<i>inv_ns</i>		<i>qtobin_s</i>		<i>qtobin_ns</i>		<i>cf_s</i>		<i>cf_ns</i>	
	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14
mean	0.127	0.108	0.165	0.089	1.011	0.763	1.376	1.155	0.143	0.110	0.144	0.121
median	0.103	0.101	0.133	0.068	0.657	0.615	1.094	0.883	0.126	0.123	0.130	0.118
maximum	0.908	0.676	0.842	0.521	6.062	3.825	7.646	7.133	0.894	0.379	0.709	0.566
minimum	0.000	0.000	0.000	0.000	0.108	0.180	0.001	0.039	-0.125	-0.652	-0.991	-0.824
standard deviation	0.112	0.077	0.143	0.085	0.908	0.508	1.061	0.899	0.107	0.099	0.184	0.117
number of observations	177	232	286	441	160	211	255	434	183	231	359	454
	<i>debto_s</i>		<i>debto_ns</i>		<i>dch_s</i>		<i>dch_ns</i>		<i>lev_s</i>		<i>lev_ns</i>	
	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14	00-08	09-14
mean	0.010	0.020	0.034	0.016	0.057	0.067	0.127	0.054	0.209	0.318	0.241	0.300
median	0.003	0.006	0.008	0.001	0.035	0.033	0.055	0.013	0.170	0.276	0.215	0.278
maximum	0.179	0.636	0.660	0.480	0.841	1.748	1.565	1.352	0.905	1.203	0.809	2.226
minimum	-0.115	-0.113	-0.213	-0.270	-0.241	-0.191	-0.434	-0.488	0.000	0.000	0.000	0.000
standard deviation	0.034	0.075	0.083	0.064	0.119	0.196	0.228	0.174	0.175	0.230	0.175	0.268
number of observations	165	216	165	216	166	216	316	466	218	232	384	460

Tab. 4. Estimation results, equation (5)

	2000-2008		2009-2014	
$inv_{i,t-1}$	0.351*		0.184*	
$qtobin_{i,t-1}$	0.079*		0.006*	
$qtobin_{i,t-1} \times state_{i,t}$	0.035*	0.114*	0.041*	0.047*
$cf_{i,t-1}$	-0.020		0.091*	
$cf_{i,t-1} \times state_{i,t}$	-0.066	-0.066	-0.164*	-0.073*
$debto_{i,t}$	0.117*		0.339*	
$debto_{i,t} \times state_{i,t}$	0.008	0.117*	-0.232*	0.107*
Number of observations	159		548	
Probability(J-statistic)	0.476		0.337	

* significant at 1% (5%, 10%)

